arr dep

900 910 → 1

940 1200 → 1

– 950 1120 → 2

– 1100 1130 → 3

– 1500 1900 → 1

1800 2000 → 2

I/P: ~~H.~~ H.
~~1~~ → 2 → 3 → 4 → N.
t

O/P: H.
2 → 3 → 4 → N.

temp = H.
H = H→next
– delete (temp)

TC: O(1).
AS: O(1).

~~H.~~ H. H.
~~1~~ → N. NULL
t t

I/P: H.
1 → 2 → 3 → 4 → N.

O/P: H.
1 → 2 → 3 → N.

H. t. t2
1 → 2 → 3 → ~~4~~ → N.
↳ N.

t2 = t → next.
t → next = N.
delete (t2).

H.
1 → 2 → 3 → 4 → N.
~~t~~ ~~t~~ t

2nd last node → temp → next → next = NULL.

TC: O(n).
AS: O(1).

H.
1 → N.
t

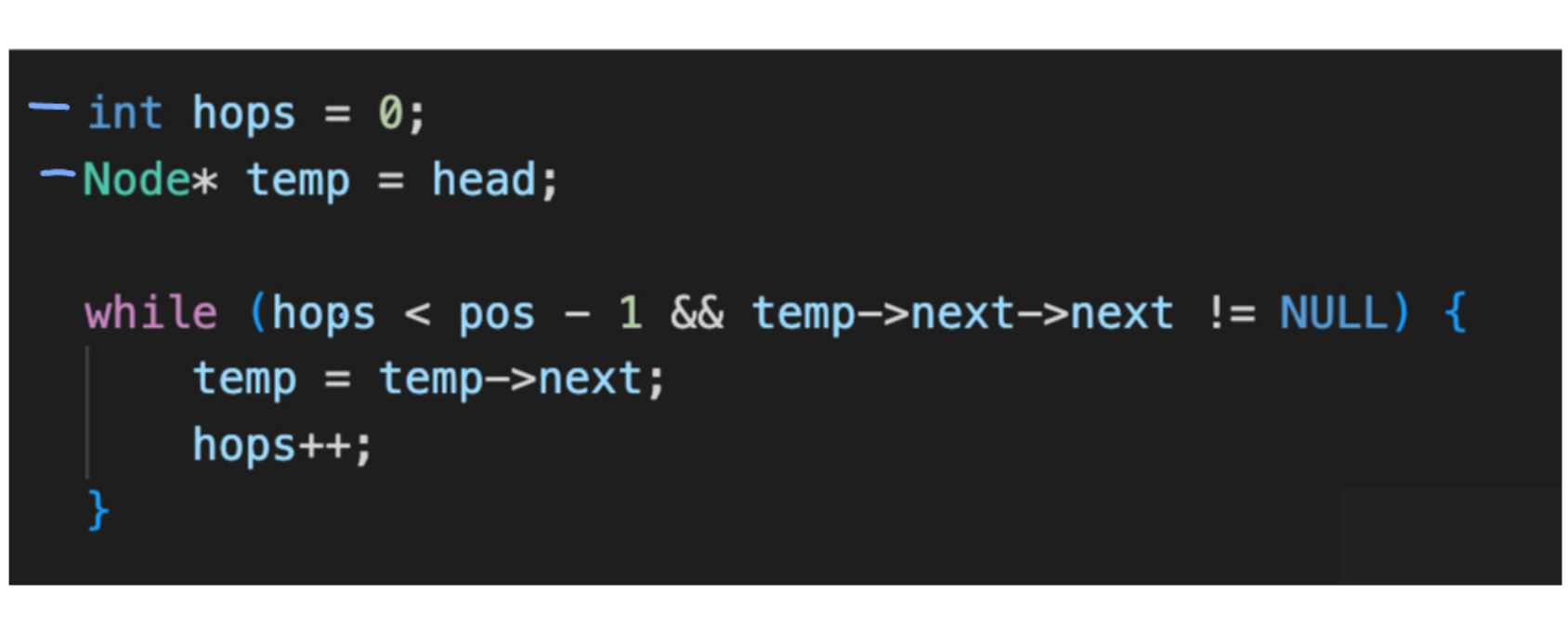
I/P: H.
1 → 2 → 3 → 4 → 5 → N. , pos = 2

O/P: H.
1 → 2 → 4 → 5 → N.

H. t. t2
1 → 2 → ~~3~~ → 4 → 5 → N.

1) Do (pos−1) hops to reach [2]
2). t2 = t → next
3) t → next = t → next → next
4) delete (t2).

TC: O (min (n, pos)).
AS: O(1).



```
int hops = 0;
Node* temp = head;

while (hops < pos - 1 && temp->next->next != NULL) {
    temp = temp->next;
    hops++;
}
```

H.
1 → 2 → 3 → ~~4~~ → N.
~~t~~ ~~t~~ t t2

pos = 100
hops = ~~0~~ ~~1~~ 2

Reverse a linked list.

I/P: H
1 → 2 → 3 → 4 → 5 → N.

O/P: H.
N ← 1 ← 2 ← 3 ← 4 ← 5

I/P: ~~prev~~ ~~curr~~ t
N ← 1 ← 2 ← 3 → 4 → 5 → N.
prev. curr.

O/P: prev. curr
N ← 1 ← 2 ← 3 4 → 5 → N.

t = curr → next
curr → next = prev.
prev = curr
curr = temp.

~~prev~~ ~~curr~~ ~~t~~ ~~t~~ ~~t~~ ~~t~~ t
N ← 1 ← 2 ← 3 ← 4 ← 5 → N.
~~p~~ ~~c~~ ~~c~~ ~~c~~ ~~c~~ c
~~p~~ ~~p~~ ~~p~~ p

newHead = prev.

TC: O(n).
AS: O(1).

Middle.

I/P: 1 → 2 → 3 → 4 → 5 → N.
O/P: 3.

I/P: 1 → 2 → 3 → 4 → 5 → 6 → N.
O/P: 4.

Runner's Technique.

ODD.
~~s~~ ~~s~~ s
1 → 2 → 3 → 4 → 5 → N.
~~f~~ ~~f~~ f

f → next == NULL.

EVEN.
~~s~~ ~~s~~ ~~s~~ s
1 → 2 → 3 → 4 → 5 → 6 → N.
~~f~~ ~~f~~ ~~f~~ f

fast == NULL.

(fast != NULL && fast → next != NULL)

~~s~~ ~~s~~ s
1 → 2 → 3 → 4 → 5 → N.
~~f~~ ~~f~~ f

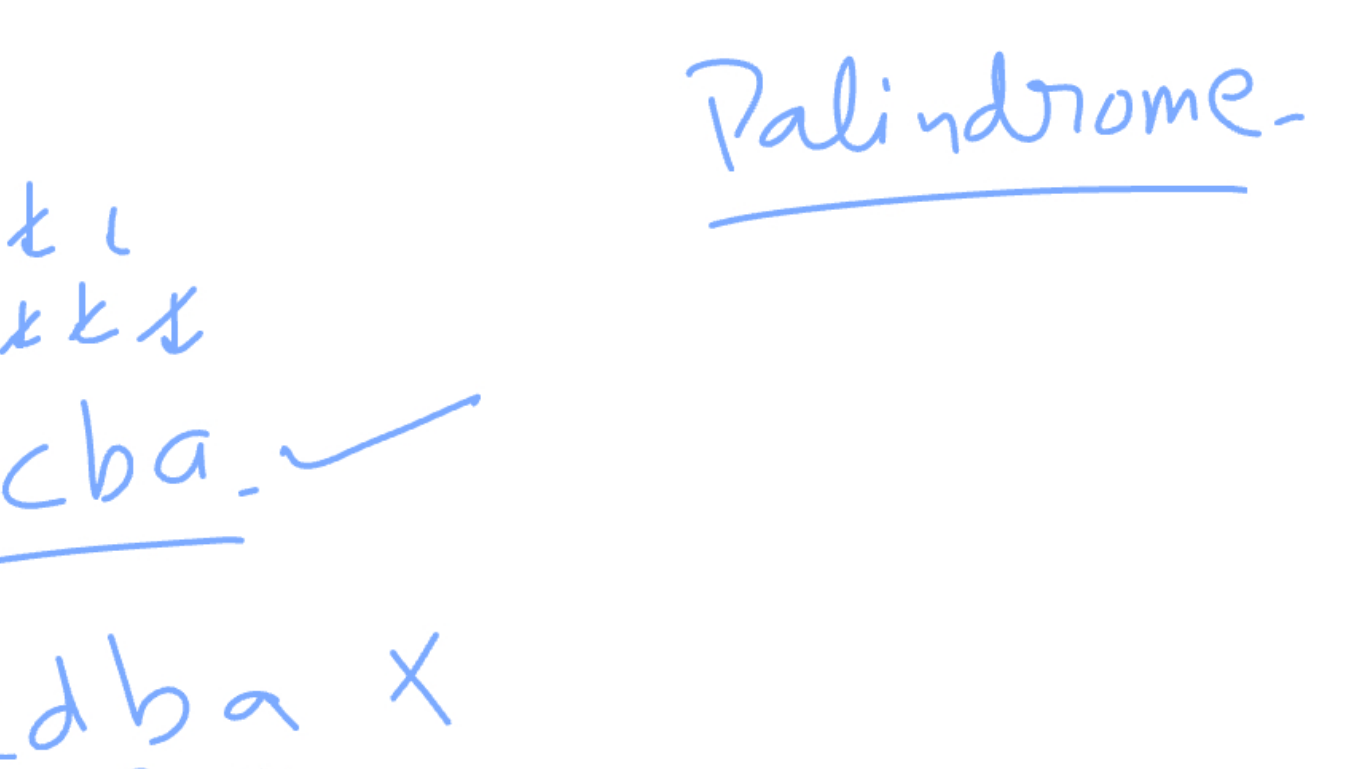
~~s~~ ~~s~~ ~~s~~ s
1 → 2 → 3 → 4 → 5 → 6 → N.
~~f~~ ~~f~~ f

TC: O(n)
AS: O(1).

i: 0 → n−1
a++

i: 0 → n−1
b++

i: 0 → n−1
– a++
– b++



```
Node* slow = head;
Node* fast = head;

while (fast != NULL && fast->next != NULL) {
    slow = slow->next;
    fast = fast->next->next;
}

return slow->data;
```

s
1 → N.
f

s
N.
f

Palindrome.

abcba. ✓

abcdba X

1 → 2 → 3 → 2 → 1 → N. ✓

1 → 2 → 2 → 1 → N. ✓

1 → 2 → 3 → 1 → N. X